Geometric Approach to Symmetric-Key Cryptanalysis

对称密码分析的几何方法

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密码学高端系列培训之十四 · 中国密码学会
北京，2026 年 8 月 25 日

Part I

Distinguisher of ciphers: property and approximation

Contents

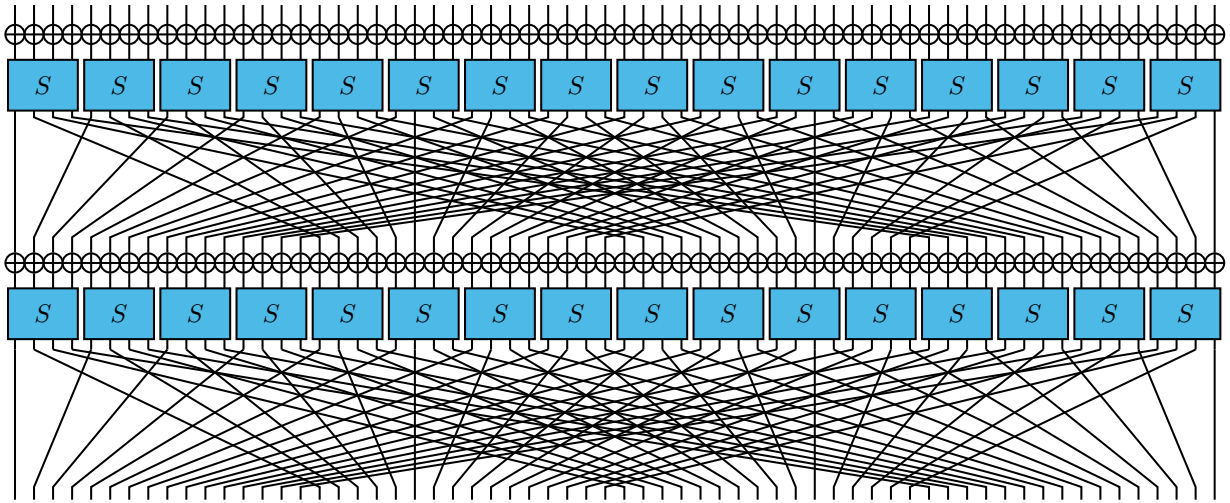
Block cipher

A block cipher maps fixed-size input blocks to fixed-size output blocks.

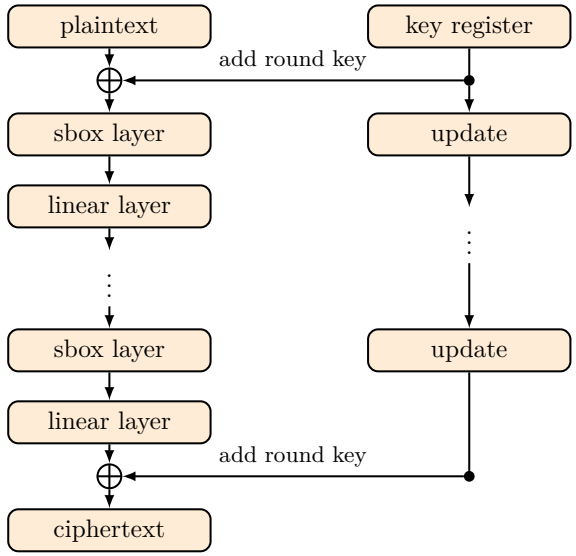
Structurally, it is typically built upon three core components:

- Non-linear layers (e.g., Sboxes for confusion)
- Linear layers (e.g., Bit-permutations for diffusion)
- Key mixing layer (e.g., Key-XORs)

Example: PRESENT block cipher



Round Structure



Overall Framework

[Bogdanov et al., CHES 2007]

Security foundations: public-key cryptosystems vs. block ciphers

Public-key cryptosystems (e.g., RSA)

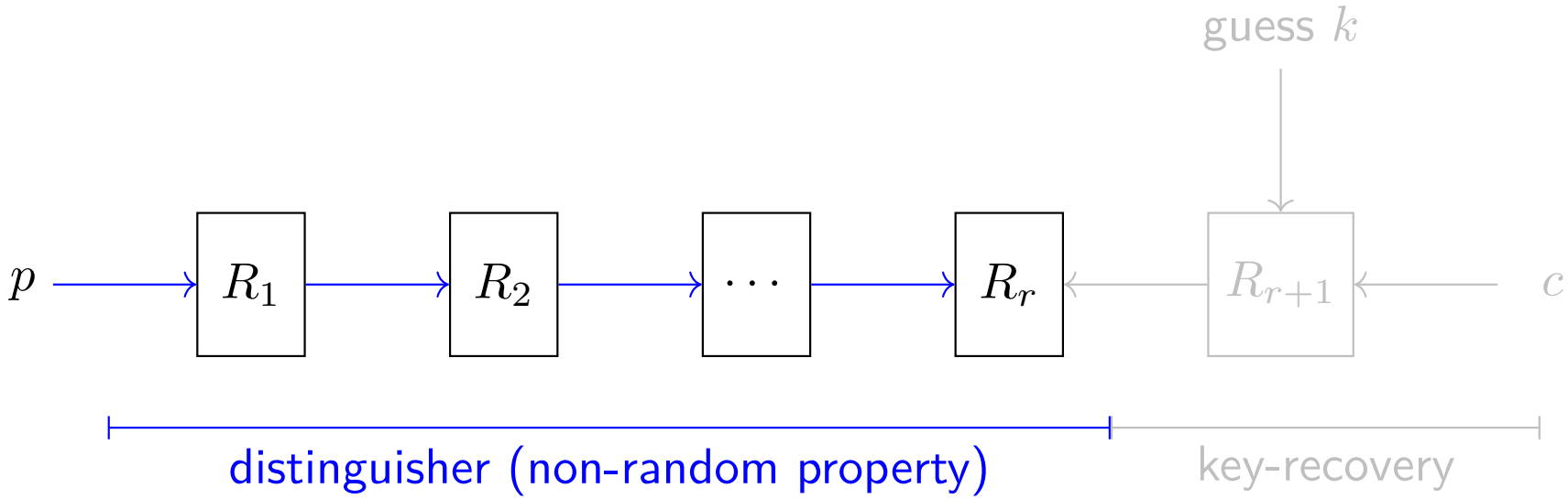
- Security relies on the hardness of specific mathematical problems.
- Example: Integer factorization.
- *Status*: No classical polynomial-time algorithm is known, but impossibility is not proven.

Block ciphers (e.g., AES, DES)

- No underlying mathematical hard problem has been identified.
- Security is based on heuristic arguments and extensive cryptanalysis.
- *Principle*: "Security through scrutiny" — if no efficient attack is found after years of analysis by experts, we consider it secure.

Cryptanalysis of block ciphers

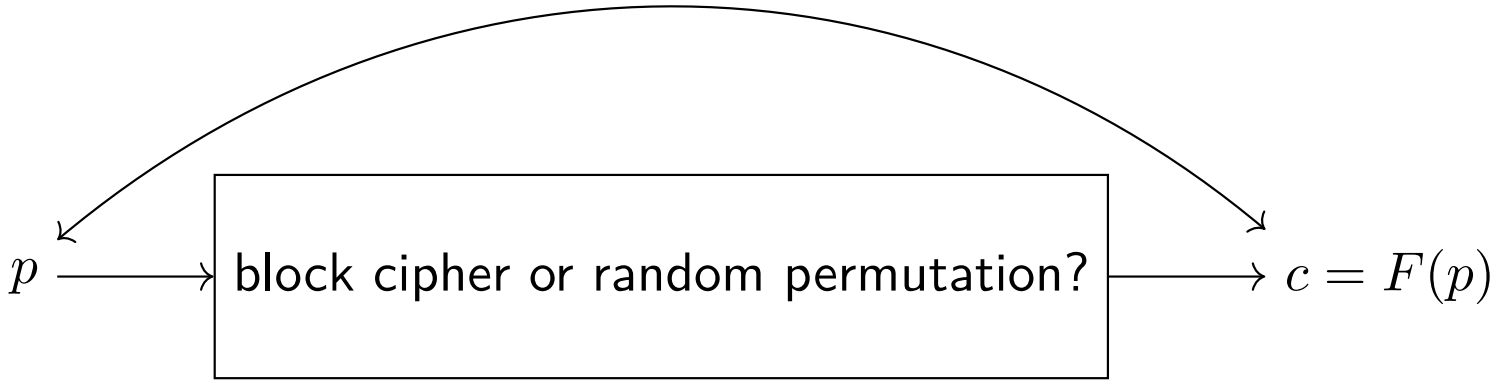
- Attackers try to distinguish it from a random permutation (i.e., an ideal cipher) using a data complexity below 2^n, i.e. without querying the full codebook of an n-bit block cipher.
- Designers try to prove that it cannot be distinguished within this bound.



Linear distinguisher

- Motivation: the input and output might be linearly related.

$$\varepsilon = \mathrm{Pro}_{p\in\mathbb{F}_2^n}\left[p_{i_1} \oplus p_{i_2} \oplus \cdots \oplus p_{i_s} = c_{j_1} \oplus c_{j_2} \oplus \cdots \oplus c_{j_t}\right] - \frac{1}{2}$$



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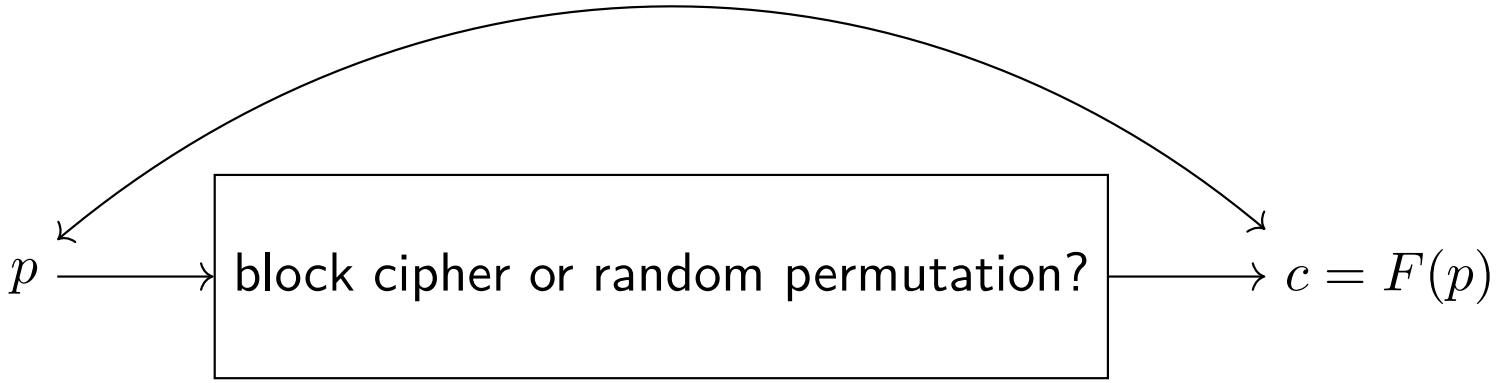

- For random: $|\varepsilon| \approx 0$
- Mask: $\gamma \in \mathbb{F}_2^n$

$$\gamma^\top x := \bigoplus_{i=0}^{n-1} \gamma_i \cdot x_i = \bigoplus_{0\le i<n, \gamma_i=1} x_i$$

- Bias:

$$\mathrm{Bias}_F(\gamma,\eta) = \Pr_{p\in\mathbb{F}_2^n}[\gamma^\top p = \eta^\top F(p)] - \frac{1}{2}$$

[Matsui, EUROCRYPT 1993]

Correlation

- Correlation of a variable $x \in \mathbb{F}_2$:

$$\begin{aligned}\mathsf{Cor}_x &= \mathrm{Pro}_x[x=0] - \mathrm{Pro}_x[x=1] \\ &= \mathrm{Pro}_x[x=0] - (1 - \mathrm{Pro}_x[x=0]) \;=\; 2\mathsf{Bias}_x\end{aligned}$$

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- Correlation represents the same thing as the bias, but is usually more convenient than bias:
  - a more convenient sum expression (see later)

$$\mathrm{Cor}_F(\gamma,\eta) = 2^{-n} \sum_{p\in\mathbb{F}_2^n} (-1)^{\gamma^\top p \oplus \eta^\top F(p)}$$

  - easier for Piling-up lemma: Let $x_0, x_1 \in \mathbb{F}_2$ be independent

$$\mathrm{Cor}_{x_0\oplus x_1} = \mathrm{Cor}_{x_0} \cdot \mathrm{Cor}_{x_1}$$

$$(\mathrm{Bias}_{x_0\oplus x_1} = 2 \cdot \mathrm{Bias}_{x_0} \cdot \mathrm{Bias}_{x_1})$$

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  - Let $X_i = (-1)^{\gamma^\top p_i \oplus \eta^\top F(p_i)}$; $E(X_i) = \theta$ and $E(X_i^2) = 1$.
  - $\mathrm{E}(T) = \theta$, $\mathrm{Var}(T) = \frac{1-\theta^2}{N} \approx \frac{1}{N}$ (assume θ is small).
  - When N is large, $T \sim \mathcal{N}(\theta, \frac{1}{N})$.

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- Random:
  - $T \sim \mathcal{N}(0, \frac{1}{N})$.

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- Connect the sign to the mask value: write $b_i = \gamma^\top p_i \oplus \eta^\top F(p_i)$, then

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$$P(X_i = +1) - P(X_i = -1) = P(b_i = 0) - P(b_i = 1) = \mathrm{Cor}_F(\gamma, \eta) = \theta$$

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- Check: $\theta = 0$ (random) gives $\mathrm{Var}(X_i) = 1$ — a fair ± 1.

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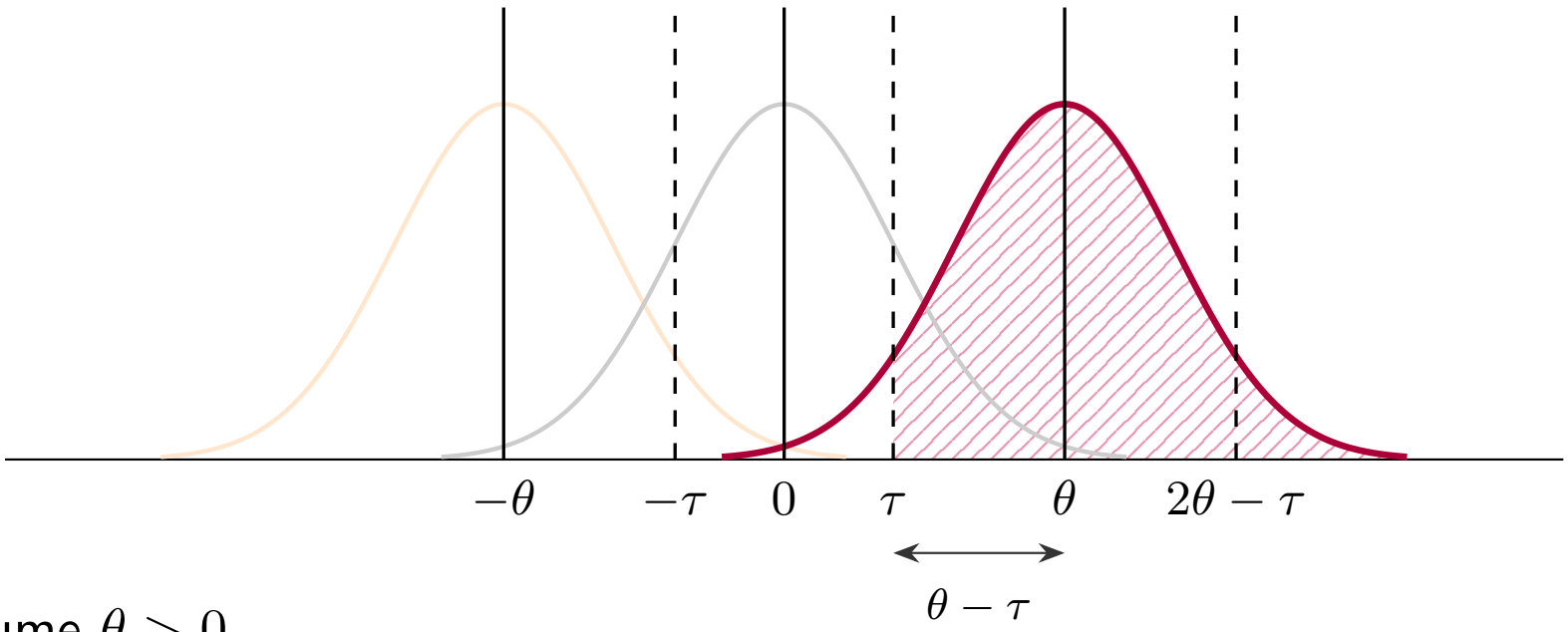
- Central limit theorem (N large): T is approximately normal,

$$T \sim \mathcal{N}\left(\theta, \tfrac{1-\theta^2}{N}\right), \qquad \tfrac{1-\theta^2}{N} \approx \tfrac{1}{N} \quad (\theta \ll 1)$$

  so $T \sim \mathcal{N}\left(\theta, \frac{1}{N}\right)$; for a random permutation ($\theta = 0$): $T \sim \mathcal{N}(0, \frac{1}{N})$.
- This is why N shows up in the data-complexity formula.

Distinguish two normal distributions (1)

- Hypothesis testing: $H_0 : T \sim \mathcal{N}(0, \frac{1}{N}); H_1 : T \sim \mathcal{N}(\theta, \frac{1}{N})$.

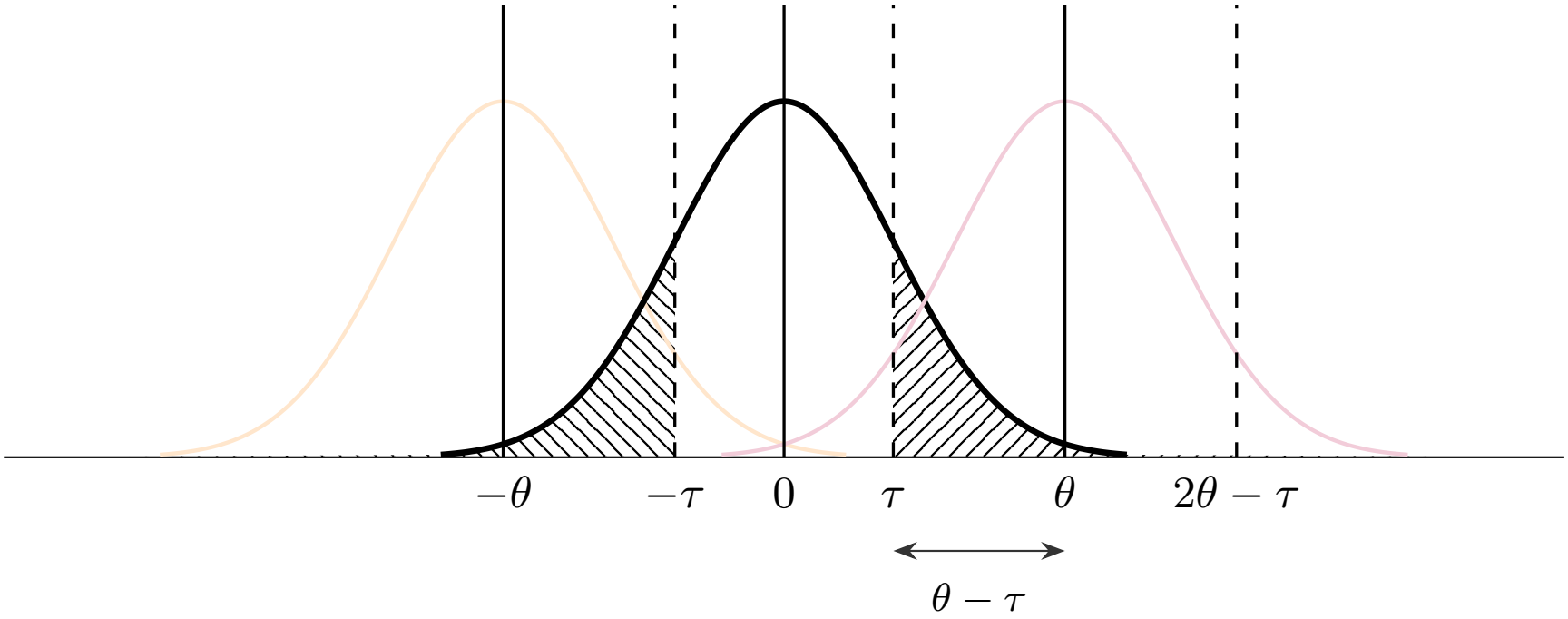


- Assume $\theta > 0$,
  - Decision rule: fix a threshold τ and declare cipher if $|T| > \tau$.
  - True positive:

$$P_S = \Phi(\sqrt{N}(\theta - \tau)) + \Phi(-\sqrt{N}(\theta + \tau)) \approx \Phi(\sqrt{N}(\theta - \tau))$$

    - Second term: $\Pr[T < -\tau]$, of order $\Phi(-\sqrt{N}\theta)$ — negligible.

Distinguish two normal distributions (2)



- False positive: $P_F = 2(1 - \Phi(\sqrt{N}\tau))$.
- Cancel τ: $\theta - \frac{\Phi^{-1}(P_S)}{\sqrt{N}} = \frac{\Phi^{-1}(1-\frac{P_F}{2})}{\sqrt{N}}$.
- Data complexity is determined by fixing P_S and P_F.

$$N = \left(\frac{\Phi^{-1}(P_S) + \Phi^{-1}(1 - \frac{P_F}{2})}{\theta}\right)^2$$

How P_S is computed

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$$\Pr[T > \tau] = \Pr[Z > \sqrt{N}(\tau - \theta)] = \Phi\big(\sqrt{N}(\theta - \tau)\big),$$

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- Hence

$$P_S = \Phi\big(\sqrt{N}(\theta - \tau)\big) + \Phi\big(-\sqrt{N}(\theta + \tau)\big),$$

with the second term $\leq \Phi(-\sqrt{N}\theta) \to 0$ (negligible).

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- Equate the two τ's and solve:

$$\theta - \frac{\Phi^{-1}(P_S)}{\sqrt{N}} = \frac{\Phi^{-1}(1 - \frac{P_F}{2})}{\sqrt{N}} \implies N = \left(\frac{\Phi^{-1}(P_S) + \Phi^{-1}(1 - \frac{P_F}{2})}{\theta}\right)^2.$$

Contents

From intuition-based approaches to systematic methods

- Different attacks have been derived from various intuitive ideas.
- Grasping the essence of these intuitions is difficult.
- Different cryptanalytic techniques appear to lack a strong theoretical connection.
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- Grasping the essence of these intuitions is difficult.
- Different cryptanalytic techniques appear to lack a strong theoretical connection.
- This makes it difficult to propose new cryptanalytic attacks.
- The geometric approach is a theoretical framework proposed by Dr. Tim Beyne to unify all cryptanalytic attacks under one coherent model.

[Beyne, ASIACRYPT 2021] [Beyne, PhD thesis, KU Leuven 2023]

Free vector spaces

- Free Vector Space: X is a finite set and $\mathbb{K}$ is a field,

$$\mathbb{K}[X] = \left\{ \sum_{x \in X} c_x \delta_x \;\middle|\; c_x \in \mathbb{K} \right\}$$

  - $x \in X \implies$ unique basis vector δ_x
  - Example: $X = \{a, b\} \implies \mathbb{F}_2[X] = \{0\,, \delta_a\,, \delta_b\,, \delta_a + \delta_b\}$

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- $\mathbb{K}[X]$ is a vector space,
  - Operations component-wise on c_x:

$$\sum_{x \in X} c_x \delta_x + \sum_{x \in X} d_x \delta_x = \sum_{x \in X} (c_x + d_x) \delta_x\,, \quad k \sum_{x \in X} c_x \delta_x = \sum_{x \in X} k c_x \delta_x$$

  - $\dim(\mathbb{K}[X]) = |X|$, and $\{\delta_x \mid x \in X\}$ is a natural standard basis.

Lifting non-linear functions to linear operators

- Intuition: study a nonlinear $F : X \to X$ using linear algebra.
- Define a linear map,

$$T^F : \mathbb{K}[X] \longrightarrow \mathbb{K}[X]; \quad \delta_x \mapsto \delta_{F(x)}$$

- Thus,

$$T^F \left(\sum_{x \in X} c_x \delta_x \right) = \sum_{x \in X} c_x \, T^F(\delta_x) = \sum_{x \in X} c_x \, \delta_{F(x)}$$

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- All linear functionals on V form a vector space (add and scale pointwise) — the dual space $V^* = \mathrm{Hom}(V, \mathbb{K})$.
- For a basis $(\beta_u, u \in X)$ of $V = \mathbb{K}[X]$, there is a unique dual basis $(\beta^u, u \in X)$ of V^*, fixed by

$$\beta^u(\beta_v) = \delta_{u,v} := \begin{cases} 1 & u = v \\ 0 & u \neq v. \end{cases}$$

  - Subscript: a basis vector. Superscript: the dual functional.

Functionals are functions $X \to \mathbb{K}$

- A functional $f : \mathbb{K}[X] \to \mathbb{K}$ is determined by its values on the standard basis $(\delta_x, x \in X)$: for any $w = \sum_x c_x \, \delta_x$,

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- So each dual functional β^u becomes a function on X, written

$$\beta^u(x) \;:=\; \beta^u(\delta_x).$$

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$$w = \sum_{u \in X} c_u \, \beta_u.$$
- Apply β^v to both sides; by $\beta^v(\beta_u) = \delta_{v,u}$, only the v-th term survives:
$$\beta^v(w) = c_v.$$

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- The dual basis $(\beta^u, u \in X)$ reads off coordinates. Write any vector w as a linear combination:
$$w = \sum_{u \in X} c_u \, \beta_u.$$
- Apply β^v to both sides; by $\beta^v(\beta_u) = \delta_{v,u}$, only the v-th term survives:
$$\beta^v(w) = c_v.$$
- Hence
$$w = \sum_{u \in X} \beta^u(w) \, \beta_u,$$
i.e. the u-th coordinate of w under $(\beta_u, u \in X)$ is the number $\beta^u(w)$.

Matrix entries via the dual basis

- The matrix of T^F under (β_u) is built column by column: its u-th column holds the coordinates of $T^F \beta_u$. So A^F is defined by

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- Coordinates with respect to a basis are unique, so (1) and (2) must agree coefficient by coefficient:

$$A^F[v,u] = \beta^v \left(T^F \beta_u\right) .$$

How to compute $A^F[v,u] = \beta^v(T^F\beta_u)$

- Goal: turn the boxed formula into a sum we can evaluate.

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- Step 3 — apply the linear functional β^v:

$$A^F[v,u] = \beta^v(T^F\beta_u) = \sum_{x\in X} \beta_u[x]\;\beta^v(\delta_{F(x)}) = \sum_{x\in X} \beta_u[x]\;\beta^v(F(x)),$$

using $\beta^v(y) := \beta^v(\delta_y)$ from the dual-basis slide.

Worked example: $X = \mathbb{F}_2$, $F =$ swap

- Standard basis $\beta_u = \delta_u$, swap $F(0) = 1$, $F(1) = 0$. So $\beta_0[0] = 1$, $\beta_0[1] = 0$, $\beta^1(0) = 0$, $\beta^1(1) = 1$.

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- One entry, $A^F[1,0] = \sum_{x \in \{0,1\}} \beta_0[x]\, \beta^1(F(x))$:

$$\beta_0[0]\,\beta^1(F(0)) + \beta_0[1]\,\beta^1(F(1)) = \beta_0[0]\,\beta^1(1) + \beta_0[1]\,\beta^1(0) = 1 \cdot 1 + 0 \cdot 0 = 1.$$

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- All four entries:

$$A^F = \begin{bmatrix} A^F[0,0] & A^F[0,1] \\ A^F[1,0] & A^F[1,1] \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Matrix representation

- Once we fix a basis for $\mathbb{K}[X]$,

$$T^F : \mathbb{K}[X] \to \mathbb{K}[X]$$

has a unique corresponding matrix, called transition matrix: it is $A^F[v,u] = \beta^v(T^F\beta_u)$ from the previous slide.

- Under the standard basis $B = (\delta_x, x \in X)$,

$$T^F[v,u] = \delta_v(F(u)) := \begin{cases} 1 & v = F(u) \\ 0 & v \neq F(u) \end{cases}$$

Example: T^F for PRESENT's Sbox (1)

- PRESENT's Sbox is a permutation $S : \mathbb{F}_2^4 \to \mathbb{F}_2^4$,

Input x	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
Output $S(x)$	c	5	6	b	9	0	a	d	3	e	f	8	4	7	1	2

$$T^S = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

Example: T^F for PRESENT's Sbox (2)

$$T^S\delta_0 = \begin{bmatrix}
0&0&0&0&0&\mathbf{1}&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&\mathbf{1}&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&\mathbf{1}\\
0&0&0&0&0&0&0&\mathbf{1}&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&\mathbf{1}&0&0&0\\
0&\mathbf{1}&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&\mathbf{1}&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&\mathbf{1}&0&0\\
0&0&0&0&0&0&0&0&0&0&0&\mathbf{1}&0&0&0&0\\
0&0&0&0&\mathbf{1}&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&\mathbf{1}&0&0&0&0&0&0&0&0&0\\
0&0&0&\mathbf{1}&0&0&0&0&0&0&0&0&0&0&0&0\\
\mathbf{1}&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&\mathbf{1}&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&\mathbf{1}&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&\mathbf{1}&0&0&0&0&0&0
\end{bmatrix} \cdot \begin{bmatrix}
\mathbf{1}\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0
\end{bmatrix} = \begin{bmatrix}
0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\\mathbf{1}\\0\\0
\end{bmatrix} = \delta_c$$

Why change the basis?

- Under the standard basis, T^F is just a permutation matrix: each entry is 0 or 1 and only records "F maps u to v".
  - All the information is there, but no cryptanalytic structure is visible.

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$\Rightarrow$ Choosing a basis = choosing a type of cryptanalysis.

[Hu et al., CRYPTO 2025]

Basis and the change-of-basis matrix

- A new basis $(\beta_x, x \in X)$ of $\mathbb{K}[X]$, written against the standard basis:

$$(\beta_x, x \in X) = (\delta_x, x \in X)\, H$$

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- $(\beta_x, x \in X)$ is a basis $\iff$ H is invertible.

A small example: $X = \mathbb{F}_2$

- $\mathbb{K}[X]$ is 2-dimensional with standard basis δ_0, δ_1. Take

$$\beta_0 = \delta_0 + \delta_1 \,, \qquad \beta_1 = \delta_0 - \delta_1 \,.$$

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- Collecting the coordinates column-wise,

$$H = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \,, \qquad H^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} H \,.$$

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- Note $\beta_u = \left[(-1)^{u^\top x}, x \in \mathbb{F}_2\right]$: this is already the linear basis, here for $n = 1$.

How coordinates transform

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So the coordinates of w under the new basis are

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! The basis vectors are transformed by H, the coordinates by H^{-1} — opposite directions. Mixing these two up is the most common source of confusion here.

How the matrix transforms

- Given $T^F u = v$ under the standard basis, substitute $u = Hu'$ and $v = Hv'$:

$$T^F H u' = H v' \implies \big(H^{-1} T^F H\big) u' = v' \,.$$

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- Transition matrix under the new basis:

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- Entry-wise this is exactly the dual-basis reading from before:

$$A^F[v, u] = \beta^v \left(T^F \beta_u\right),$$

since the v-th row of H^{-1} is β^v and the u-th column of H is β_u.

A coordinate of A^F in closed form

- Reading off one entry of $A^F = H^{-1}T^F H$: the v-th row of H^{-1} is β^v, the u-th column of H is β_u:

$$\begin{aligned}
A^F[v,u] &= \left[\beta^v(a),\, a \in \mathbb{F}_2^n\right]^\top \cdot T^F \cdot \left[\beta_u[a],\, a \in \mathbb{F}_2^n\right] \\
&= \left[\sum_a \beta^v(a) T^F_{a,0}, \sum_a \beta^v(a) T^F_{a,1}, \ldots, \sum_a \beta^v(a) T^F_{a,2^n-1}\right]^\top \\
&\qquad \cdot \left[\beta_u[a],\, a \in \mathbb{F}_2^n\right] \\
&= \left[\beta^v(F(x)), x \in \mathbb{F}_2^n\right]^\top \cdot \left[\beta_u[a],\, a \in \mathbb{F}_2^n\right] \\
&= \sum_{x \in \mathbb{F}_2^n} \beta^v(F(x))\beta_u[x]
\end{aligned}$$

From matrix coordinates to distinguishers

- Fix a pair (v, u) and suppose the entry behaves differently in the two worlds:

  $A^F[v, u] = \theta \neq 0$ for the cipher, $\quad A^F[v, u] = 0$ for a random permutation.

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  $A^F[v,u] = \theta \neq 0$ for the cipher, $\quad A^F[v,u] = 0$ for a random permutation.

- Suppose moreover that $A^F[v,u]$ can be estimated from data: that it is the mean of N independent ± 1 observations. Then we are exactly back in the first part, and the data complexity is

$$N = \left(\frac{\Phi^{-1}(P_S) + \Phi^{-1}(1 - \frac{P_F}{2})}{\theta} \right)^2 .$$

Example-1: change-of-basis with the linear basis

- For linear cryptanalysis, $\mathbb{K} := \mathbb{R}$, $X = \mathbb{F}_2^n$.
- Linear basis: $H = (\beta_u, u \in \mathbb{F}_2^n)$ where

$$\beta_u = \left[(-1)^{u^\top x}, x \in \mathbb{F}_2^n\right]$$

Then, $H^{-1} = 2^{-n} H$

$$C^F = H^{-1} T^F H$$

- We have

$$\begin{aligned} C^F[v,u] &= 2^{-n}[(-1)^{v^\top x}, x \in \mathbb{F}_2^n]^\top \cdot T^F \cdot [(-1)^{u^\top x}, x \in \mathbb{F}_2^n] \\ &= 2^{-n}[(-1)^{v^\top F(x)}, x \in \mathbb{F}_2^n]^\top \cdot [(-1)^{u^\top x}, x \in \mathbb{F}_2^n] \\ &= 2^{-n} \sum_{x \in \mathbb{F}_2^n} (-1)^{u^\top x \oplus v^\top F(x)} \end{aligned}$$

Example-1: apply linear basis to PRESENT's Sbox

$$C^F = H^{-1} \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0
\end{bmatrix} H$$

Linear approximation table of PRESENT's Sbox

- Linear approximation table (LAT):

$$C^F = \frac{1}{16}\begin{bmatrix}
16 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 8 & 0 & -8 & 0 & 8 & 0 & 8 \\
0 & 0 & 4 & 4 & -4 & -4 & 0 & 0 & 4 & -4 & 8 & 0 & 0 & 8 & 4 & -4 \\
0 & 0 & 4 & 4 & 4 & 4 & -8 & 8 & -4 & -4 & 0 & 0 & 0 & 0 & 4 & 4 \\
0 & 0 & -4 & 4 & -4 & -4 & 0 & 8 & 0 & 0 & 4 & -4 & -4 & -4 & -8 & 0 \\
0 & -8 & -4 & -4 & -4 & 4 & 0 & 0 & 0 & 0 & 4 & -4 & -4 & -4 & 8 & 0 \\
0 & 0 & 0 & -8 & 0 & 0 & -8 & 0 & -4 & 4 & 4 & 4 & -4 & 4 & -4 & -4 \\
0 & -8 & 0 & 0 & 8 & 0 & 0 & 0 & 4 & -4 & -4 & -4 & -4 & 4 & -4 & -4 \\
0 & 0 & 4 & -4 & -4 & 4 & 0 & 0 & -4 & -4 & 0 & -8 & 8 & 0 & -4 & -4 \\
0 & 0 & -4 & 4 & -4 & 4 & -8 & -8 & 4 & -4 & 0 & 0 & 0 & 0 & -4 & 4 \\
0 & 0 & 0 & -8 & 0 & -8 & 0 & 0 & 0 & -8 & 0 & 0 & 0 & 0 & 0 & 8 \\
0 & 0 & 8 & 0 & -8 & 0 & 0 & 0 & 0 & 0 & -8 & 0 & -8 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 8 & 8 & 0 & -4 & -4 & 4 & 4 & -4 & 4 & -4 & 4 \\
0 & -8 & 8 & 0 & 0 & 0 & 0 & 0 & 4 & 4 & 4 & 4 & 4 & -4 & -4 & 4 \\
0 & 0 & -4 & -4 & -4 & 4 & 0 & 8 & 8 & 0 & -4 & 4 & 4 & 4 & 0 & 0 \\
0 & 8 & 4 & -4 & 4 & 4 & 0 & 0 & 8 & 0 & 4 & -4 & -4 & -4 & 0 & 0
\end{bmatrix}$$

Contents

Approximation

- Impossible to compute the exact value of a matrix coordinate (complexity is at least $\mathcal{O}(2^n)$)!
- Approximation is necessary!
- For efficient approximation, various assumptions are necessary!

Composition

- Under the standard basis,

  For $F = F_2 \circ F_1$, where $F_i : \mathbb{F}_2^n \to \mathbb{F}_2^n$ for $i = 1, 2$

$$T^F = T^{F_2} \cdot T^{F_1}$$

  Proof:

$$T^F \delta_x = \delta_{F(x)} = \delta_{F_2(F_1(x))} = T^{F_2} \delta_{F_1(x)} = T^{F_2} T^{F_1} \delta_x.$$

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- Under any basis,

  A similar theorem applies to the matrix under a different basis

  $$A^F = A^{F_2} \cdot A^{F_1}$$

  Proof:

  $$A^F = H^{-1} T^F H = (H^{-1} T^{F_2} H)(H^{-1} T^{F_1} H) = A^{F_2} \cdot A^{F_1}.$$

Parallel functions

- Under the standard basis, for $F = F_2||F_1$,

$$T^F = T^{F_2} \otimes T^{F_1}$$

Proof:

$$T^F(\delta_x \otimes \delta_y) = \delta_{F_2(x)} \otimes \delta_{F_1(y)} = (T^{F_2} \otimes T^{F_1})(\delta_x \otimes \delta_y)$$

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- Under any basis: on the product space the natural basis is $\beta_u \otimes \beta_v$, with change-of-basis matrix $H \otimes H$, so

$$A^F = (H \otimes H)^{-1} T^F (H \otimes H) = (H^{-1} \otimes H^{-1}) T^F (H \otimes H)\,.$$

Parallel functions

- Under the standard basis, for $F = F_2 || F_1$,

$$T^F = T^{F_2} \otimes T^{F_1}$$

Proof:

$$T^F(\delta_x \otimes \delta_y) = \delta_{F_2(x)} \otimes \delta_{F_1(y)} = (T^{F_2} \otimes T^{F_1})(\delta_x \otimes \delta_y)$$

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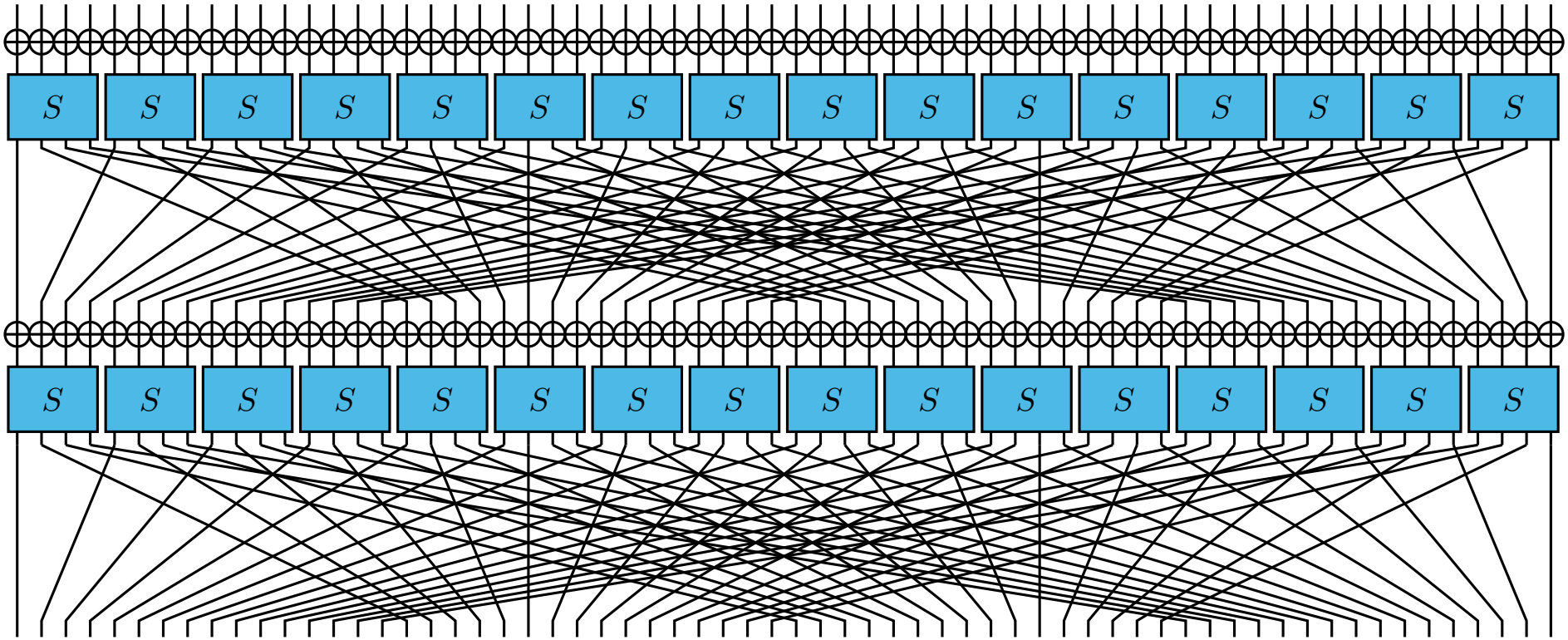
With this, $A^F = A^{F_2} \otimes A^{F_1}$ follows:

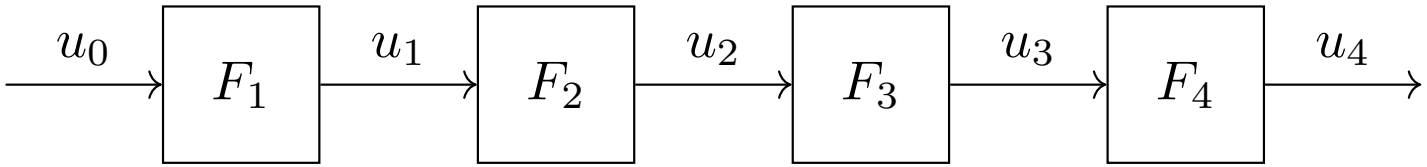
$$\begin{aligned} A^{F_2} \otimes A^{F_1} &= (H^{-1} T^{F_2} H) \otimes (H^{-1} T^{F_1} H) \\ &= (H^{-1} \otimes H^{-1})(T^{F_2} \otimes T^{F_1})(H \otimes H) \\ &= (H^{-1} \otimes H^{-1}) T^F (H \otimes H) \;=\; A^F\,. \end{aligned}$$

Apply the geometric approach to a full cipher



- Three types of components: Key-XOR, Sbox, and Linear layer.
- Construct matrices for each component.

Express matrix coordinate



- Composition:

$$A^F = A^{F_r \circ \cdots \circ F_1} = A^{F_r} \cdots A^{F_2} \cdot A^{F_1}$$

$$A^F[u_r, u_0] = \sum_{u_1, \ldots, u_{r-1}} A^{F_r}[u_r, u_{r-1}] \cdot A^{F_{r-1}}[u_{r-1}, u_{r-2}]$$
$$\cdots A^{F_2}[u_2, u_1] \cdot A^{F_1}[u_1, u_0]$$

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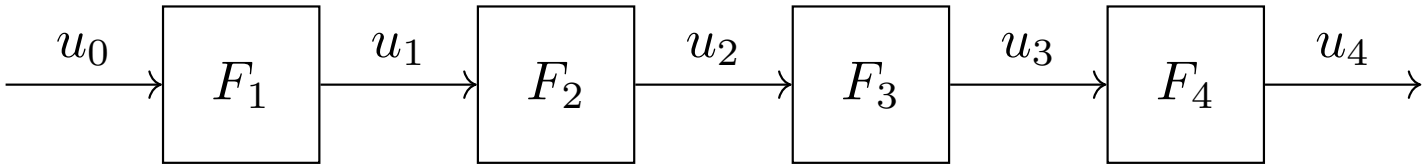

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- A trail:

$$(u_0, u_1, \ldots, u_r)$$

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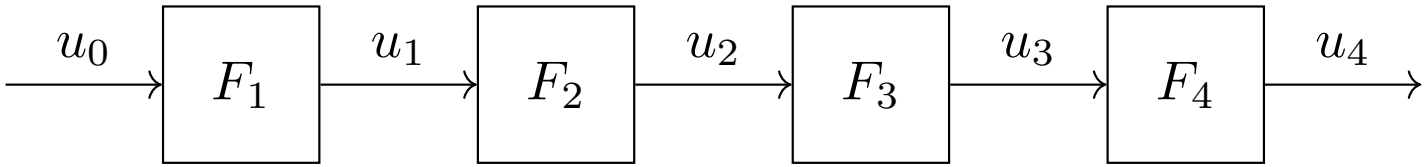

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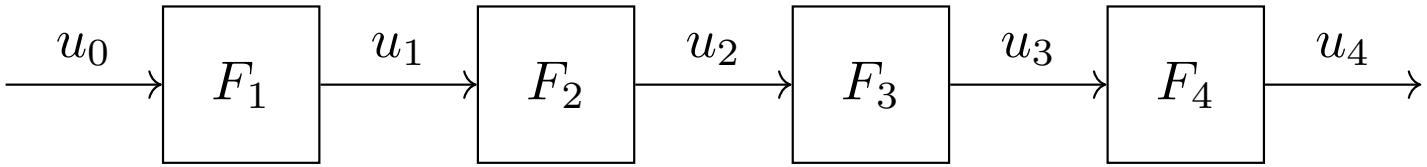

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- Result:

$A^F[u_r, u_0]$ is the sum of all trails' correlations !

Approximation in linear cryptanalysis (1)

- From here on we fix the linear basis, so every matrix is written C instead of the generic A: $C^F = H^{-1}T^F H$ with H the linear basis, and $C^F[v,u]$ is a correlation.
- Linear trail: $(u_0, u_1, \ldots, u_r)$ (aka linear characteristic).
- Dominant trail assumption:

$$\begin{aligned} C^F[u_r,u_0] &= \sum_{u_1,\ldots,u_{r-1}} C^{F_r}[u_r,u_{r-1}]\cdot C^{F_{r-1}}[u_{r-1},u_{r-2}] \\ &\qquad \cdots C^{F_2}[u_2,u_1]\cdot C^{F_1}[u_1,u_0] \\ &\approx C^{F_r}[u_r,u_{r-1}]\cdot C^{F_{r-1}}[u_{r-1},u_{r-2}]\cdots C^{F_2}[u_2,u_1] \\ &\qquad \cdots C^{F_1}[u_1,u_0] \end{aligned}$$

Approximation in linear cryptanalysis (2)

- In linear cryptanalysis, we assume

$$|C^F[u_r, u_0]| \approx \max_{u_1,\ldots,u_{r-1}} \Big| C^{F_r}[u_r, u_{r-1}] \cdot C^{F_{r-1}}[u_{r-1}, u_{r-2}] \cdots C^{F_2}[u_2, u_1] \cdot C^{F_1}[u_1, u_0] \Big|$$

- Need to find out a linear trail with the maximum absolute correlation.

Approximation in linear cryptanalysis (3)

- Key-XOR (fixed-key): $F_k : \mathbb{F}_2^n \to \mathbb{F}_2^n ; x \mapsto x \oplus k$

$$C^{F_k}[v,u] = (-1)^{v^\top k}\delta_v(u)$$

- Sbox: $S : \mathbb{F}_2^m \to \mathbb{F}_2^m$, $C^S[v,u]$ from LAT
- Linear layer:

$$C^F[v,u] = \delta_u(L^\top v)$$

- A trail:

$$u_0 \underbrace{\xrightarrow{\text{Key-XOR}}}_{(-1)^{u_1^\top k}, u_0 = u_1} u_1 \underbrace{\xrightarrow{\text{Sbox}}}_{C^S[u_2,u_1]} u_2 \underbrace{\xrightarrow{\text{Linear layer } L}}_{L^\top u_3 = u_2} u_3$$

Finding the best trail

- The task: among all trails $(u_0, u_1, \ldots, u_r)$, find one maximising $\left|\prod_i C^{F_i}[u_i, u_{i-1}]\right|$, i.e. minimising the weight

$$w = \sum_{i=1}^{r} -\log_2 \left|C^{F_i}[u_i, u_{i-1}]\right| .$$

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- Every constraint comes directly from what we already derived:
  - Sbox: the non-zero entries of the LAT are the feasible transitions, each with weight $-\log_2 |C^S[v, u]|$.
  - Linear layer: $u = L^\top v$; for a bit-permutation, just a rewiring.
  - Key-XOR: $C^{F_k}[v, u] = (-1)^{v^\top k}\delta_v(u)$ only flips a sign, so it does not change $|C|$ — the search is key-independent.

Result: best linear trails of round-reduced PRESENT

Solving that model for PRESENT's Sbox and bit-permutation layer:

rounds r	minimal weight	$\lvert C\rvert$ of the best trail
1	1	2^{-1}
2	2	2^{-2}
3	4	2^{-4}
4	6	2^{-6}

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A best 4-round trail — exactly one active Sbox per round:

round	input mask	output mask	weight
1	0a00000000000000	0200000000000000	1
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3	0080000000000000	0080000000000000	2
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Careful: this is the correlation of one trail. The true $C^F[u_r, u_0]$ is the sum over all trails — the dominant trail assumption is exactly what we are relying on.

Part I takeaway

- A distinguisher is a number that behaves differently for the cipher and for a random permutation. Its size θ fixes the data complexity,

$$N = \left((\Phi^{-1}(P_S) + \Phi^{-1}(1 - \tfrac{P_F}{2}))/\theta\right)^2.$$

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$\Rightarrow$ Finding the best trail is a combinatorial optimisation problem, handed to an MILP/SAT solver.

Part II

Round-based approximation of differential-linear correlation

Contents

Differential-linear distinguisher

- In Part I one sample was one message. A differential-linear (DL) distinguisher uses a pair: fix an input difference Δ and an output mask λ, and measure

$$\mathsf{DL}[\Delta \xrightarrow{F} \lambda] \;=\; \mathsf{Cor}_x\left[\,\lambda^\top\big(F(x)\oplus F(x\oplus\Delta)\big)\,\right] \;=\; 2^{-n}\sum_{x\in\mathbb{F}_2^n}(-1)^{\lambda^\top D_\Delta F(x)},$$

  where $D_\Delta F(x) = F(x)\oplus F(x\oplus\Delta)$ is the derivative of F in the direction Δ.

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  where $D_\Delta F(x) = F(x)\oplus F(x\oplus\Delta)$ is the derivative of F in the direction Δ.

- For a random permutation this is ≈ 0. So it is again a statistic that separates the two worlds — and the analysis of Part I applies verbatim, now with $\theta = \mathrm{DL}[\Delta \to \lambda]$.

[Langford, Hellman, CRYPTO 1994]

One sample is a pair: the 2-wise form

- The statistic involves x and $x \oplus \Delta$ together, so the object we push through the cipher is a pair. Define the 2-wise form

$$F^{\times 2} : \mathbb{F}_2^n \times \mathbb{F}_2^n \longrightarrow \mathbb{F}_2^n \times \mathbb{F}_2^n ; \qquad (x, \Delta) \longmapsto \big(F(x),\ D_\Delta F(x)\big) .$$

  - The second coordinate is the difference, not $F(x \oplus \Delta)$: the same information, but it keeps Δ in plain sight.

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- The wise d of an attack is the number of messages in one sample:

$$d = 1 \text{ (linear, Part I)}, \qquad d = 2 \text{ (DL)}, \qquad d = 2^k \text{ (higher-order DL)}.$$

The matrix of $F^{\times 2}$: no new machinery

- $F^{\times 2}$ acts on $\mathbb{F}_2^n \times \mathbb{F}_2^n$, so its basis is the Kronecker product of two copies of the linear basis — exactly the situation of *Parallel functions* in Part I:

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$\Rightarrow$ We can again work round by round — with the same tools as Part I, only on a space of twice the dimension.

Correlation vector of a multiset

- Part I attached a correlation to a function. The same expression makes sense for a multiset S of values in $X = \mathbb{F}_2^k$, and defines the correlation vector $\mathsf{CV}(S)$:

$$\mathsf{CV}(S)[u] \;=\; \frac{1}{|S|}\sum_{x\in S}(-1)^{u^\top x}, \qquad \mathsf{CV}(S) \in \mathbb{R}[X].$$

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- $\mathrm{CV}(S)[0] = 1$ always; the remaining coordinates measure how far S is from the uniform distribution.
  - A set of plaintexts, a set of ciphertexts, a set of differences — each has one.

The correlation vector propagates by C^F

- Let $\mathcal{C} = F(\mathcal{P})$, i.e. apply F to every element of the multiset $\mathcal{P}$. Then

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$\Rightarrow$ Pushing a set of plaintexts through the cipher = multiplying its correlation vector by one correlation matrix per round.

[Daemen, Govaerts, Vandewalle, FSE 1994] [Beyne, ASIACRYPT 2021]

DL correlation is one coordinate

- Feed in all messages with one fixed difference:

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$$\mathsf{CV}(\mathcal{C})\big[(0,\lambda)\big] \;=\; 2^{-n} \sum_{x \in \mathbb{F}_2^n} (-1)^{\lambda^\top D_\Delta F(x)} \;=\; \mathsf{DL}[\Delta \xrightarrow{F} \lambda]\,.$$

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$\Rightarrow$ Computing a DL correlation = propagating one vector through the rounds and reading one coordinate at the end.

Contents

The wall: exact DL needs every trail

- Part I turned a correlation into a sum over trails, then kept the dominant one. For DL the same expansion is available and exact:

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$\Rightarrow$ We need an approximation that never enumerates trails — stay inside the geometric framework, but propagate round by round.

Round by round: two cheap steps per round

- By $C^{E^{\times 2}} = \prod_i C^{F_i^{\times 2}}$ we push the correlation vector through the rounds. Split an SPN round as $F_i = \mathsf{L} \circ \mathsf{S}$:

$$\mathsf{CV}(\mathcal{P}) = \sigma_0 \xrightarrow{C^{\mathsf{S}^{\times 2}}} \gamma_1 \xrightarrow{C^{\mathsf{L}^{\times 2}}} \sigma_1 \to \cdots \to \gamma_r = \mathsf{CV}(\mathcal{C}).$$

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  - S-box layer: a Kronecker product, so it acts S-box by S-box — exact.
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! We never build the $2^{2n} \times 2^{2n}$ matrix; only local vectors of length 2^{2m}.

Prepare the plaintext vector σ_0

- The plaintext multiset is $\mathcal{P} = \mathbb{F}_2^n \times \{\Delta\}$. By Part I's parallel rule its correlation vector factorizes across the S-boxes:

$$\sigma_0 = \mathsf{CV}(\mathcal{P}) = \bigotimes_{i=0}^{t-1} \mathsf{CV}\big(\mathbb{F}_2^m \times \{\varpi_i(\Delta)\}\big) =: \bigotimes_{i=0}^{t-1} \sigma_0^{[i]},$$

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- ϖ_i is the projection onto the i-th S-box coordinate:

$$\varpi_i : \mathbb{F}_2^{mt} \to \mathbb{F}_2^m, \qquad v \mapsto v_i.$$

Prepare the plaintext vector σ_0

- The plaintext multiset is $\mathcal{P} = \mathbb{F}_2^n \times \{\Delta\}$. By Part I's parallel rule its correlation vector factorizes across the S-boxes:

$$\sigma_0 = \mathrm{CV}(\mathcal{P}) = \bigotimes_{i=0}^{t-1} \mathrm{CV}\big(\mathbb{F}_2^m \times \{\varpi_i(\Delta)\}\big) =: \bigotimes_{i=0}^{t-1} \sigma_0^{[i]},$$

  each $\sigma_0^{[i]} \in \mathbb{R}\big[\mathbb{F}_2^{2m}\big]$.

- ϖ_i is the projection onto the i-th S-box coordinate:

$$\varpi_i : \mathbb{F}_2^{mt} \to \mathbb{F}_2^m, \qquad v \mapsto v_i.$$

- Cost: only t small vectors to build, $\mathcal{O}(t \cdot 2^m)$.

Pass the S-box layer

- The S-box layer is $\mathsf{S} = \mathsf{s} \| \cdots \| \mathsf{s}$ (t parallel copies), so by Part I's parallel rule

$$C^{\mathsf{S}^{\times 2}} = \bigotimes_{i=0}^{t-1} C^{\mathsf{s}^{\times 2}}.$$

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- Hence applying it keeps the Kronecker form, acting S-box by S-box:

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  i.e. each $\gamma_1^{[i]} = C^{\mathsf{s}^{\times 2}} \sigma_0^{[i]}$ is computed independently.

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  i.e. each $\gamma_1^{[i]} = C^{\mathsf{s}^{\times 2}} \sigma_0^{[i]}$ is computed independently.

- Local and exact — no error yet. Multiplying by one $C^{\mathsf{s}^{\times 2}}$ fast is the key trick, next.

The S-box step factorizes — the key trick

- A 2-wise S-box entry is a product of two ordinary LAT entries:

$$C^{\mathsf{s}^{\times 2}}\big[(v_0, v_1), (u_0, u_1)\big] \;=\; C^{\mathsf{s}}\big[v_0 \oplus v_1,\; u_0 \oplus u_1\big] \cdot C^{\mathsf{s}}\big[v_1,\; u_1\big].$$

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- So no new table is needed — the LAT of Part I already contains everything.
- The factorization also turns $\gamma^{[i]} = C^{\mathsf{s}^{\times 2}} \sigma^{[i]}$ into a two-step contraction (sum over u_0, then over u_1): the cost of a whole layer drops from 2^{4m} to $\mathcal{O}(t \cdot 2^{3m})$.

Pass the linear layer

- A linear layer acts on correlation vectors by mask transpose:

$$\sigma_1[(v_0, v_1)] = \gamma_1\big[(\mathsf{L}^{\times 2})^\top (v_0, v_1)\big] = \gamma_1\big[(\mathsf{L}^\top v_0,\, \mathsf{L}^\top v_1)\big].$$

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- Reading $\gamma_1 = \bigotimes_j \gamma_1^{[j]}$ at this permuted mask means projecting back to each S-box:

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- The result is generally not a Kronecker product across S-boxes anymore — this mixing is exactly what forces the approximation on the next slide.

The linear step: the only approximation

- The mixed result is generally not a Kronecker product. We force it to be one — a rank-1 approximation:

$$\sigma_i \approx \bigotimes_{j=0}^{t-1} \sigma_i^{[j]} .$$

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- The mixed result is generally not a Kronecker product. We force it to be one — a rank-1 approximation:

$$\sigma_i \;\approx\; \bigotimes_{j=0}^{t-1} \sigma_i^{[j]} \,.$$

$\Rightarrow$ Only the t local marginals $\sigma_i^{[j]} \in \mathbb{R}[\mathbb{F}_2^{2m}]$ are kept, i.e. $\mathcal{O}(t\,2^{2m})$ memory. This is the only source of error — it says the S-boxes behave independently, and here that assumption is written down explicitly rather than hidden.

The whole algorithm

Input (Δ, λ); keep local vectors $\sigma^{[j]} \in \mathbb{R}[\mathbb{F}_2^{2m}]$.

1. **Init**: $\sigma_0^{[j]} = \mathsf{CV}\big(\mathbb{F}_2^m \times \{\Delta^{[j]}\}\big)$.

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   (a) S-box: $\gamma_i^{[j]} = C^{\mathsf{s}^{\times 2}} \sigma_{i-1}^{[j]}$ (exact)
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   (c) Key / constant XOR: a sign $(-1)^{v_0^\top k}$ (next slide)

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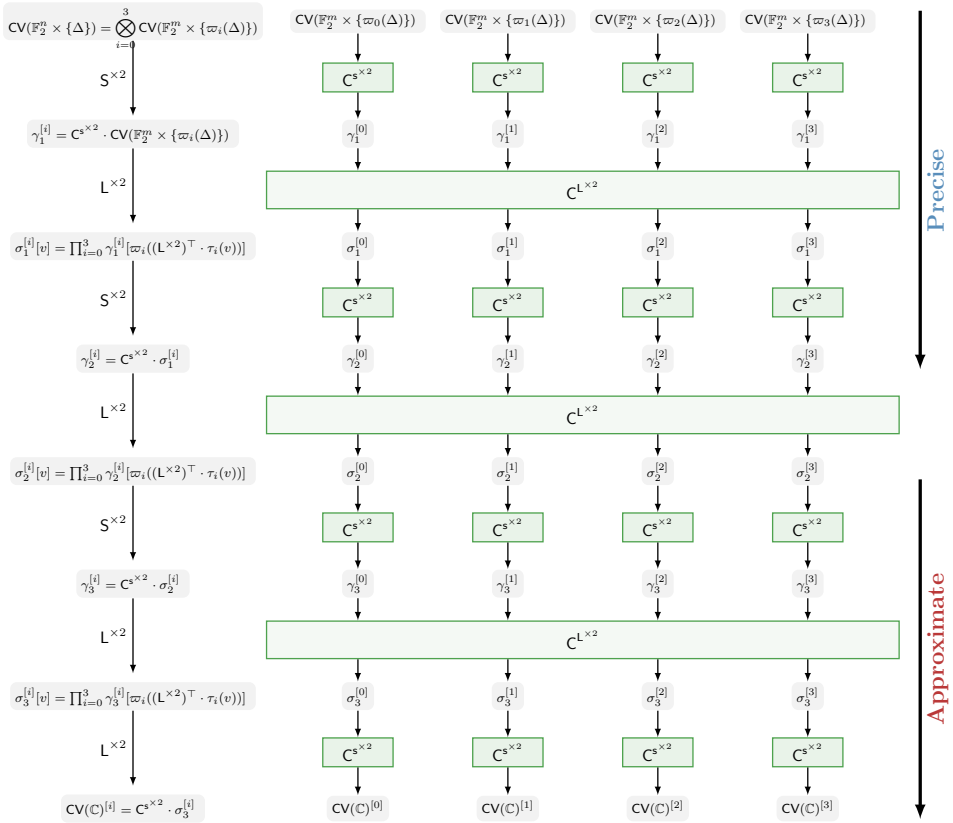
Cost $\mathcal{O}(t\, 2^{3m})$ per round; for $m = 4, 5$ this runs in milliseconds, with no trail enumeration at all.

The pipeline at a glance

Local vectors $\sigma^{[j]}, \gamma^{[j]}$ move down; each round applies $C^{\mathsf{s}\times 2}$ and then $C^{\mathsf{L}\times 2}$. Round 1 is exact; the rank-1 error enters from round 2 on.



Controlling the error: extend, then approximate

- The rank-1 assumption only bites once the difference has spread across S-boxes. So treat the early rounds differently. Split $F = F_2 \circ F_1$:

$$\mathsf{DL}[\Delta \xrightarrow{F} \lambda] = \sum_{\Delta'} p_{\Delta'} \cdot \mathsf{DL}[\Delta' \xrightarrow{F_2} \lambda], \qquad p_{\Delta'} = \Pr_x \left[D_\Delta F_1(x) = \Delta' \right].$$

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$\Rightarrow$ Postponing the moment the assumption breaks buys longer and tighter distinguishers.

Key-XOR is just a sign flip

- For $\mathsf{K}_k : (x, \theta) \mapsto (x \oplus k, \theta)$ the 2-wise correlation matrix is diagonal:

$$C^{\mathsf{K}_k^{\times 2}}\left[(v_0, v_1), (u_0, u_1)\right] = (-1)^{v_0^\top k}\, \delta_{u_0}(v_0)\, \delta_{u_1}(v_1).$$

  XORing k only multiplies coordinate (v_0, v_1) by a sign depending on the value mask v_0.

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- Average key: k uniform $\Rightarrow$ every coordinate with $v_0 \neq 0$ vanishes, so one may set $v_0 = 0$ safely.

Average key: why the value mask must vanish

- Under average key, k is uniform in $\mathbb{F}_2^n$, so the sign $(-1)^{v_0^\top k}$ is averaged over all keys:

$$\mathbb{E}_k\left[(-1)^{v_0^\top k}\right] \;=\; 2^{-n} \sum_{k \in \mathbb{F}_2^n} (-1)^{v_0^\top k}.$$

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$$\sum_{k\in\mathbb{F}_2^n} (-1)^{u^\top k} \;=\; 2^{n-1}\cdot(+1) + 2^{n-1}\cdot(-1) \;=\; 0,$$

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- Hence

$$\mathbb{E}_k\left[(-1)^{v_0^\top k}\right] \;=\; \begin{cases} 1 & v_0 = 0, \\ 0 & v_0 \neq 0, \end{cases}$$

  i.e. every coordinate with $v_0 \neq 0$ vanishes — only $v_0 = 0$ survives.

Ascon: most precise, and one more round

Ascon is a sponge-based design whose permutation is an SPN with small S-boxes, so the pipeline above applies to it unchanged.

Target	Rounds	Th. Cor.	Method
Ascon-128 init.	4	2^{-1}	DLCT / HATF / TDT
	4	2^{-1}	round-based
	5	$2^{-9.1}$	TDT
	5	$2^{-8.94}$	round-based (matches experiment)
Ascon-128a init.	6	$2^{-23.89}$	round-based (new)
Ascon-p	5	$2^{-4.0}$ / $2^{-6.83}$	GDLCT
	5	$2^{-4.21}$ / $2^{-7.58}$	round-based
	6	$-2^{-21.89}$	round-based (new)

Theory follows experiment closely, and reaches one more round than every earlier method.

PRESENT: from 13 to 17 rounds

Target	Rounds	Th. Cor.	Method
PRESENT	13	$2^{-27.01}$	GDLCT
PRESENT	13	$2^{-22.43}$	round-based
PRESENT	**17**	$2^{-29.76}$	round-based (new)

- At 13 rounds the correlation is much stronger than the earlier estimate — the earlier method was losing a lot.
- Valid DL distinguishers go from 13 to 17 rounds: a 4-round jump on the cipher we used throughout Part I.

Higher-order DL: same method, wider tuples

- Replace pairs by 2^d-tuples — the 2^d-wise form. Nothing in the pipeline changes; only the local alphabet grows.

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- For an SPN with t parallel m-bit S-boxes, a d-th order DL evaluation costs

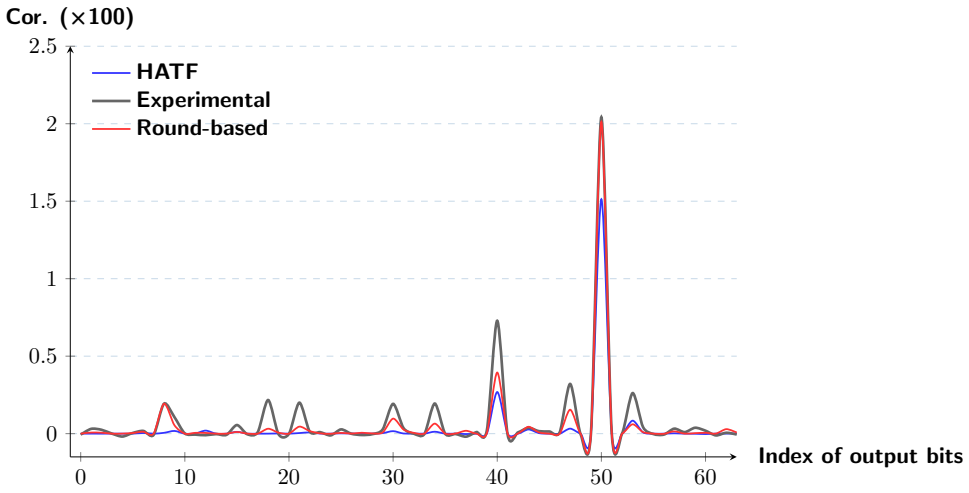
$$\mathcal{O}\big(t \cdot 2^{(2^d+1)m}\big).$$

- So $d = 2$ (quartets) is comfortably in reach for $m = 4, 5$, while $d = 3$ is already expensive.

Second-order DL on Ascon-128

Rounds	Exp.	Th. Cor.	Method
5	$2^{-5.60}$	$2^{-6.05}$	HATF
		$2^{-5.63}$	round-based
6	—	$2^{-45.35}$	round-based (new)

At 5 rounds the round-based estimate is closer to the experiment than HATF's; and it gives the first valid second-order DL on 6-round initialization.



5 rounds, all 64 single-bit output masks: round-based tracks the experiment and beats HATF.

Where we are

- DL correlation $=$ the $(0, \lambda)$ coordinate of the ciphertext correlation vector; propagate that vector round by round.
- S-box step exact (a product of two LAT entries); linear step a rank-1 approximation — the single assumption.
- $\mathcal{O}(t\, 2^{3m})$ per round, no trail enumeration; best known DL / higher-order DL for Ascon and Present; key handled exactly as a sign.

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The remaining problem: ciphers whose nonlinear layer is one huge χ on 257 bits. There are no small S-boxes to factor over — the local trick breaks down.

Contents

The problem: one huge χ

- SUBTERRANEAN 2.0 and KOALA have a single nonlinear layer: χ acting on 257 bits — not a row of small S-boxes.

The problem: one huge χ

- SUBTERRANEAN 2.0 and KOALA have a single nonlinear layer: χ acting on 257 bits — not a row of small S-boxes.
- Everything in the previous section factored over small S-boxes. With one 257-bit χ there is nothing to factor over, and we would need the correlation matrix of χ — and of $\chi^{\times 2}$ — as a whole. Even though χ is quadratic, that matrix has 2^{514} entries.

The problem: one huge χ

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$\Rightarrow$ Key idea: χ is a Mealy machine with a two-bit state. Its correlation matrix entries are products of constant-size matrices, so we never write the big matrix down.

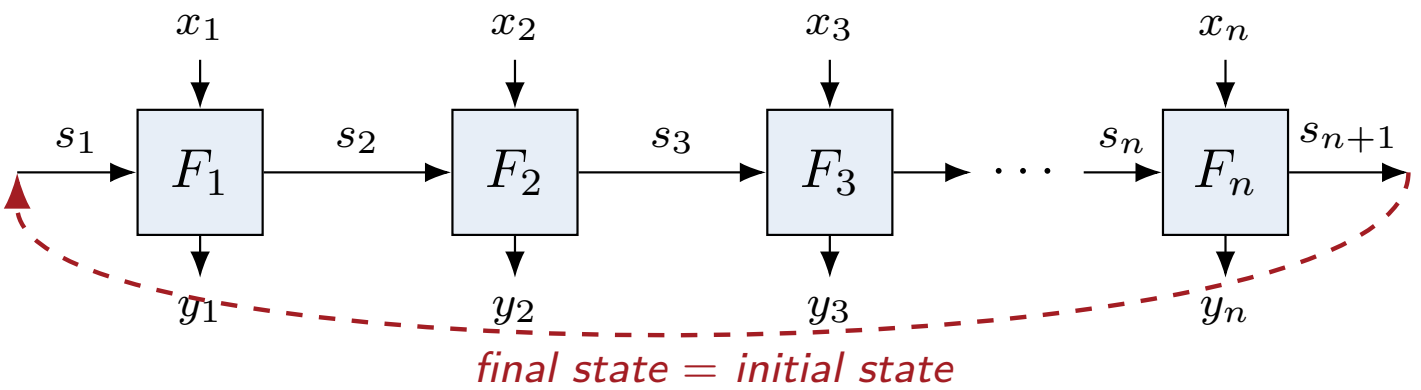
[Niu, Beyne, Hu, Wang, *Cryptanalytic Properties of Mealy Machines*]

χ is a Mealy machine (S-function)

A Mealy machine reads input chunks x_i, writes output chunks y_i, and carries a small state s_i between them. For χ,

$$y_i = x_i \oplus (x_{i-1} \oplus 1)\, x_{i-2} \quad \text{(indices cyclic)},$$

so each y_i depends on the previous two inputs: take $s_i = (x_{i-1}, x_{i-2}) \in \mathbb{F}_2^2$.



$$F_i\big((x_{i-1}, x_{i-2});\, x_i\big) = \big((x_i, x_{i-1}),\; x_i \oplus (x_{i-1} \oplus 1)x_{i-2}\big).$$

The state stays two bits even at $n = 257$.

Why the chain does not simply split

For the next three slides we drop back to a general basis: the argument works for any of them, and the linear basis of Part I is just one choice.

- Part I gives any property as a pair (u, v) with value $v(T^F u)$. Take both in tensor-product form, $u = u_1 \otimes \cdots \otimes u_n$ and $v = v_1 \otimes \cdots \otimes v_n$:

$$v(T^F u) = \sum_{x_1,\ldots,x_n} \underbrace{u_1[x_1] \cdots u_n[x_n]}_{\text{factors over } i} \cdot \underbrace{v_1(y_1) \cdots v_n(y_n)}_{\text{coupled through the state}} .$$

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! But the coupling between the past and the future passes entirely through the single state s_i. Fix s_i and the chain falls apart.

Splitting over the state: a matrix product appears

- Insert a sum over all state sequences $s_1, \ldots, s_{n+1}$ allowed by F. With the states pinned down, the product factors over i:

$$v(T^F u) = \sum_{s_1,\ldots,s_{n+1}} \prod_{i=1}^{n} \underbrace{\sum_{x_i,y_i} u_i[x_i]\, v_i(y_i)\, \big[F_i(s_i, x_i) = (s_{i+1}, y_i)\big]}_{=:\, M_i[s_{i+1}, s_i]} .$$

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- The outer sum over states is exactly a matrix product, with the initial and final state encoded by the vectors a, b:

$$M_i = \begin{bmatrix} \sum_{\substack{x_i,y_i \\ F_i(0,x_i)=(0,y_i)}} u_i[x_i]\, v_i(y_i) & \sum_{\substack{x_i,y_i \\ F_i(1,x_i)=(0,y_i)}} u_i[x_i]\, v_i(y_i) \\ \sum_{\substack{x_i,y_i \\ F_i(0,x_i)=(1,y_i)}} u_i[x_i]\, v_i(y_i) & \sum_{\substack{x_i,y_i \\ F_i(1,x_i)=(1,y_i)}} u_i[x_i]\, v_i(y_i) \end{bmatrix}$$

$x_i, y_i \in \mathbb{F}_2$; rows s_{i+1} / cols s_i

Splitting over the state: a matrix product appears

- Insert a sum over all state sequences $s_1, \dots, s_{n+1}$ allowed by F. With the states pinned down, the product factors over i:

$$v(T^F u) = \sum_{s_1,\dots,s_{n+1}} \prod_{i=1}^{n} \underbrace{\sum_{x_i,y_i} u_i[x_i]\, v_i(y_i)\, \big[F_i(s_i, x_i) = (s_{i+1}, y_i)\big]}_{=:\, M_i[s_{i+1}, s_i]} .$$

- The outer sum over states is exactly a matrix product, with the initial and final state encoded by the vectors a, b:

$$M_i = \begin{bmatrix} \sum_{\substack{x_i,y_i \\ F_i(0,x_i)=(0,y_i)}} u_i[x_i]\, v_i(y_i) & \sum_{\substack{x_i,y_i \\ F_i(1,x_i)=(0,y_i)}} u_i[x_i]\, v_i(y_i) \\ \sum_{\substack{x_i,y_i \\ F_i(0,x_i)=(1,y_i)}} u_i[x_i]\, v_i(y_i) & \sum_{\substack{x_i,y_i \\ F_i(1,x_i)=(1,y_i)}} u_i[x_i]\, v_i(y_i) \end{bmatrix}$$

$x_i, y_i \in \mathbb{F}_2$; rows s_{i+1} / cols s_i

- Each M_i has size $|S_{i+1}| \times |S_i|$ — constant, independent of n. It is nothing but Part I's change of basis applied to one Mealy step.

Choosing the basis is still choosing the attack

- The formula $v(T^F u) = b\, M_n \cdots M_1\, a$ holds for any basis vectors u_i, v_i — the machinery does not care which property we are after:
  - the linear basis of Part I $\Rightarrow$ ordinary correlation,
  - the 2-wise basis of this part $\Rightarrow$ differential-linear,
  - other choices of basis give the other classical attacks (not covered today).

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$\Rightarrow$ One theorem, evaluated with different (u_i, v_i), gives all of them — exactly the Part I slogan, now for Mealy machines.

Evaluating any property in $\mathcal{O}(n)$

Input: basis vectors $u_1, \ldots, u_n$, $v_1, \ldots, v_n$ and boundary vectors a, b. **Output**: $v(T^F u)$.

1. Transfer matrices. For $i = 1, \ldots, n$: start from $M_i = 0$ and, for every (s_i, x_i), put $(s_{i+1}, y_i) \leftarrow F_i(s_i, x_i)$ and

$$M_i[s_{i+1}, s_i] \; += \; u_i[x_i]\, v_i(y_i).$$

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Each M_i is read straight off the truth table of F_i, and the whole evaluation is n constant-size matrix-vector products: $\mathcal{O}(n)$ time, optimal in bit operations.

χ is cyclic $\Rightarrow$ the product becomes a trace

- χ feeds the final state back into the initial one, so we must force them equal: take $a = \delta_s$, $b = \delta^s$ and sum over the free state $s \in \mathbb{F}_2^2$.

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- Summing $\delta^s(M\delta_s)$ over s is precisely the trace:

$$C^{\chi}[u \to v] = \sum_{s \in \mathbb{F}_2^2} \delta^s \Big(\prod_i M_i \, \delta_s \Big) = \mathrm{Tr}\Big(\prod_{i=1}^{n} M_i \Big).$$

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- Since all F_i are the same map, in the linear basis there are only four distinct matrices M_i — one for each $(u_i, v_i) \in \mathbb{F}_2 \times \mathbb{F}_2$, each 4×4 (rows and columns indexed by the state (x_{i-1}, x_{i-2})).

From χ to $\chi^{\times 2}$

- For DL we need the 2-wise form. Nothing conceptual changes: run the same construction on pairs, i.e. on the Mealy machine $G_i = F_i \times F_i$ with state (s_i, s_i').

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- The state is now four bits, so the transfer matrices are 16×16 — still constant size — and

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$\Rightarrow$ This makes $C^{\chi^{\times 2}}$ computable for a 257-bit χ for the first time, and it replaces the bespoke recursion that earlier work needed for χ: it is one instance of the general Mealy-machine theorem.

From one entry to the whole vector

- The trace gives one entry of $C^{\chi^{\times 2}}$. But the round-based step needs the whole vector moved:

$$\gamma \;=\; C^{\chi^{\times 2}}\sigma\,, \qquad \text{i.e.} \qquad \gamma[v] \;=\; \sum_{u\in\mathbb{F}_2^n\times\mathbb{F}_2^n} C^{\chi^{\times 2}}[v,u]\;\sigma[u]\,,$$

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- Use the same rank-1 trick as for S-box ciphers, now per bit: with no small S-boxes, each single bit plays that role,

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  Here $u_j := (u_0[j], u_1[j])$ and $v_j := (v_0[j], v_1[j])$.

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$\Rightarrow$ Both factors of the sum — the trace and σ — now factor bit by bit. That is what makes the sum collapse.

Absorbing σ into the transfer matrices

- Write both factors per bit and exchange sum and product:

$$\gamma[v] = \sum_{u} \operatorname{Tr}\Big(\prod_{j} M_j(u_j, v_j)\Big) \prod_{j} \sigma^{[j]}[u_j]$$

$$= \operatorname{Tr}\Big(\prod_{j} \underbrace{\sum_{u_j} \sigma^{[j]}[u_j]\, M_j(u_j, v_j)}_{=:\ \widetilde{M}_j(v_j;\sigma^{[j]})}\Big)$$

Legal: each u_j meets one matrix factor and one σ factor; product and trace are linear.

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  Legal: each u_j meets one matrix factor and one σ factor; product and trace are linear.

- Each $\widetilde{M}_j$ is again 16×16 — constant size — so

$$\gamma[v] \;=\; \operatorname{Tr}\Big(\widetilde{M}_{n-1} \cdots \widetilde{M}_0\Big), \qquad \widetilde{M}_j = \widetilde{M}_j\big(v_j;\, \sigma^{[j]}\big).$$

  ! $\widetilde{M}_j$ is not intrinsic to χ: it carries the incoming $\sigma^{[j]}$, so it is rebuilt every round.

  $\Rightarrow$ n constant-size products and one trace: $\mathcal{O}(n)$, with no mask enumeration.

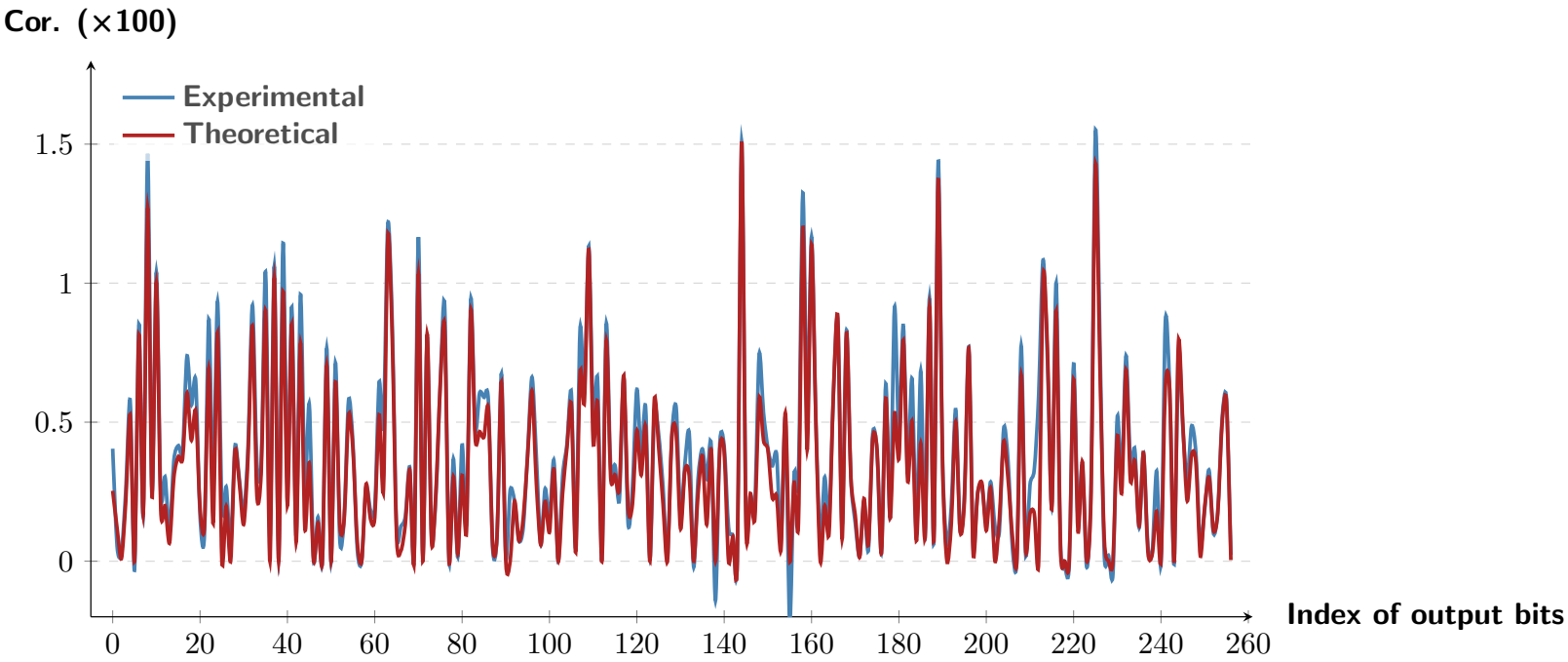
Results for Subterranean and Koala

Target	Rounds	Th. Cor.	Note
Subterranean (DL)	5	$-2^{-7.98}$	exp. $-2^{-7.88}$
	6	$2^{-20.09}$	new
Koala-p (DL)	5	$2^{-6.87}$	exp. $2^{-6.73}$
	6	$2^{-21.42}$	new
Subterranean (2nd order)	5	$2^{-6.05}$	matches exp.
	7	$2^{-90.99}$	first
Koala-p (2nd)	7	$2^{-86.24}$	first

DL distinguishers reach two rounds beyond the best differential or linear ones, and these are the first 7-round second-order DL distinguishers for both permutations.

How precise? 5-round SUBTERRANEAN, all 257 masks

Second-order DL: theory against experiment for all 257 single-bit output masks.



The two curves are visually indistinguishable — the estimate stays sharp even with one giant nonlinear layer. KOALA-p behaves the same way.

Part II takeaway

- A DL correlation is one coordinate of a correlation vector; propagating that vector round by round replaces trail enumeration entirely.
- The S-box step is exact (a product of two LAT entries); the linear step is a rank-1 approximation — the single, explicit assumption.
- A key XOR is a sign on the value mask, so fixed-key estimates are sound.
- When the nonlinear layer is one huge χ, view it as a Mealy machine: every property becomes a product (here a trace) of constant-size matrices, evaluated in $\mathcal{O}(n)$.

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Same principle throughout: choose a basis, then read one coordinate.

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